IBM Z DATATHON 2026 · WILDCARD: BEYOND THE OBVIOUS

YouthQuest AI

An AI-personalised, gamified youth development platform.
Level up your life by levelling up your community.

Volunteering · Social projects · Real-life rewards · Early-warning AI · Powered by IBM LinuxONE

THE PROBLEM

Youth potential is wasted, social ills fill the gap

Rising social problems

Drug abuse, vaping, gangsterism, online gambling and digital addiction pull youth in.

Fragmented programs

Volunteer and social projects are one-size-fits-all; youth lose interest fast.

No recognition or path

Contributions go untracked and unrewarded; no visible growth journey.

Late intervention

NGOs, schools and government lack real-time data on who is disengaging.

OUR SOLUTION

Make doing good more rewarding than doing harm

1 · AI quests

Personalised volunteering, projects, courses and mentoring matched to each youth.

2 · Verified impact

QR + GPS + photo check-in with AI fraud detection.

3 · Gamification

XP, levels, badges, streaks, team challenges, national leaderboards.

4 · Real rewards

Vouchers, transit passes, certificates, internships, scholarships from CSR partners.

5 · Early warning

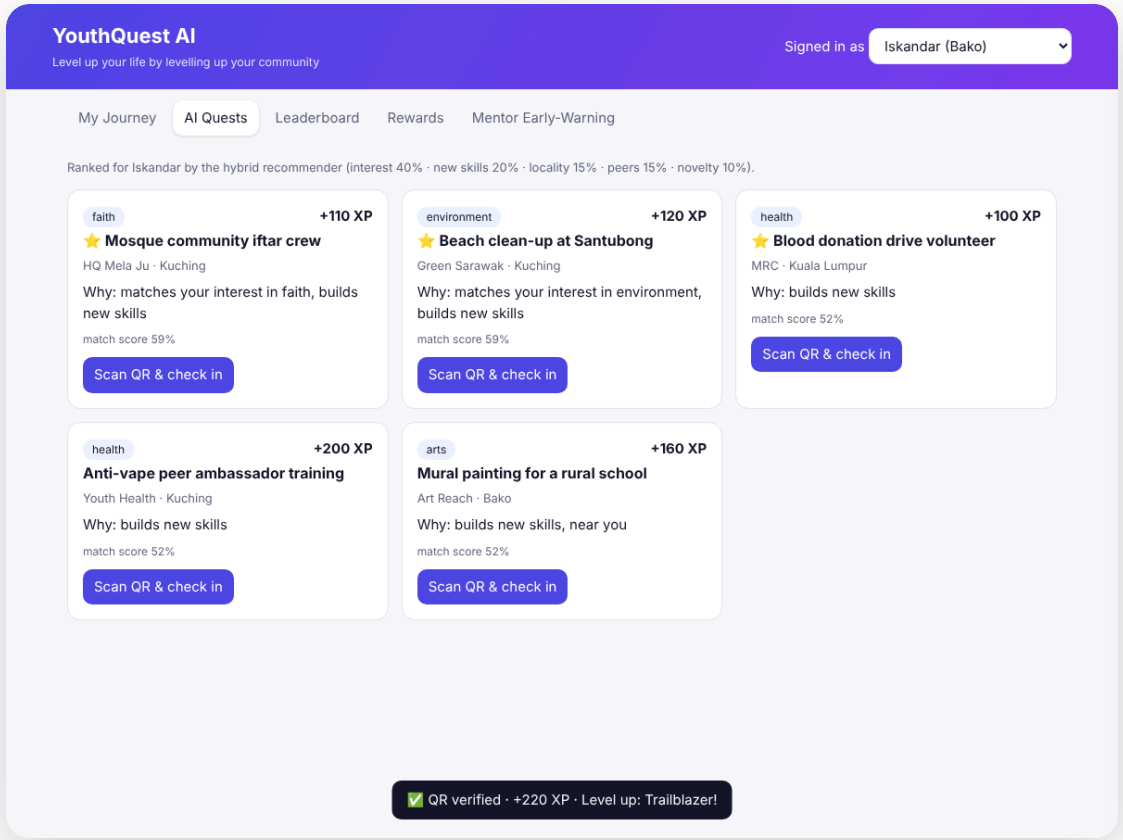
Risk model flags disengaging youths so mentors step in early.

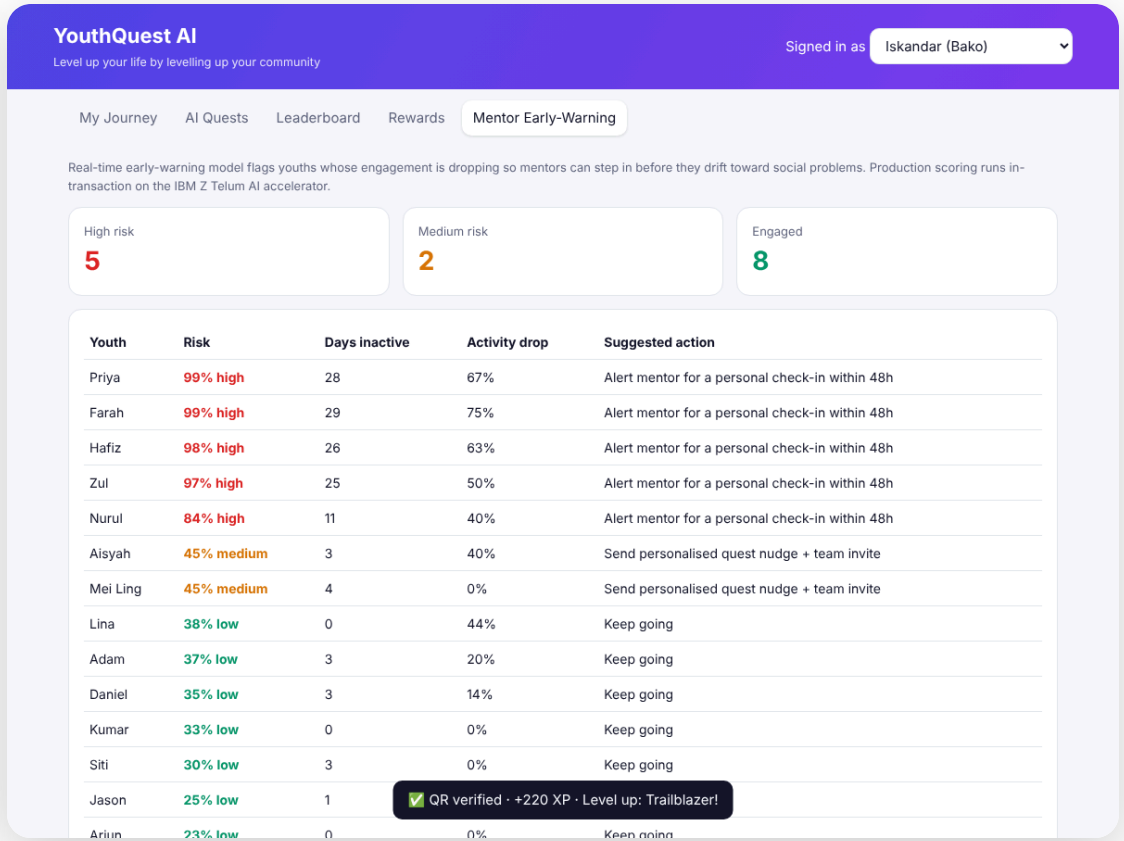
6 · LLM coach

Personal growth guidance and reflection after each quest.

WORKING PROTOTYPE

Live demo





Left: AI-ranked quests with explanations and one-tap QR check-in. Right: mentor early-warning dashboard with suggested actions.

HOW THE AI WORKS

Personalised, explainable, real-time

Hybrid recommender

Score = 40% interest + 20% new skills + 15% locality + 15% peer signal + 10% novelty, with a plain-language reason.

Early-warning risk model

Logistic model on days inactive, activity drop, broken streak and interest diversity → high / medium / low with action.

Tamper-evident ledger

Hash-chained, idempotent points ledger; duplicate scans ignored, no double-spend on rewards.

Stack

Python / FastAPI · Db2 / PostgreSQL · PyTorch + Snap ML · React Native · Red Hat OpenShift.

IBM Z / LINUXONE

Why it belongs on IBM Z

Protect youth data

Pervasive encryption, Secure Execution and Crypto Express HSM for minors' data and PDPA compliance.

Real-time AI

Telum on-chip AI accelerator scores risk and recommendations in-transaction at check-in.

Fraud-proof rewards

Db2 ACID transactions back the points ledger and redemptions.

National scale, green

OpenShift on LinuxONE autoscales to millions of users; consolidation lowers energy use.

IMPACT

Who benefits

Youth 15–30
Growth path, skills, real rewards, kept away from social ills.

NGOs
Steady pipeline of motivated, verified volunteers.

Schools & unis
Track co-curricular growth and impact.

Government
Real-time youth engagement data and early intervention.

Sponsors
Transparent CSR impact and a talent pipeline.

↑ volunteer retention · M+ verified volunteer hours · ↓ youth risk behaviour

ROADMAP

From prototype to national scale

Phase 1 · Pilot

Deploy on LinuxONE Community Cloud with 3 universities and 10 NGOs in Sarawak.

Phase 2 · Scale

Train models on real engagement data; onboard CSR reward partners; mobile app.

Phase 3 · National

Integrate with government youth agencies; state & national leagues.

The largest youth development ecosystem ever, one quest at a time.